

point in this landscape can be assigned a fitness score equal to Q . We can think of the Louvain algorithm as moving through this landscape, looking for solutions of increasingly greater fitness (highly-modular partitions). If the landscape features a single highly-fit partition (meaning that this partition is much more modular than other nearby partitions), we expect the Louvain algorithm to successfully negotiate the landscape and, more times than not, return as its output a partition that approximates the optimal partition. However, if the landscape features multiple near-optimal partitions we expect the Louvain algorithm to have difficulty arriving at the same solution each run. We refer to this type of landscape as a *degenerate* or *near-degenerate* modularity landscape, indicating the presence of multiple near-optimal solutions (Good et al., 2010).

With this intuition in mind, we calculated the similarity of all pairs of detected partitions at time t (a total of $100(100 - 1)/2 = 4950$ pairs) using the z-score of the Rand index (Traud et al., 2011) and subsequently averaged over all pairs, thus obtaining a single score, $Z(t)$, that quantified the average pairwise similarity of partitions at time t . If $Z(t)$ was large, it indicated the the Louvain algorithm consistently returned similar partitions. On the other hand, if $Z(t)$ was small, then the partitions varied from run to run. We interpreted the magnitude of $Z(t)$ as an indication of the level of degeneracy in the modularity landscape, with high values indicating low levels of degeneracy. It should be noted that the Louvain algorithm uses a greedy heuristic to maximize modularity and very likely samples near-optimal partitions in a somewhat biased manner.